

**A STUDY ON CONSENSUS MONTE CARLO:
METROPOLIS–HASTINGS ON DISTRIBUTED DATA**

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FINAL YEAR PROJECT REPORT

**A STUDY ON CONSENSUS MONTE CARLO:
METROPOLIS–HASTINGS ON DISTRIBUTED DATA**

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DECLARATION

I hereby declare that this thesis is based on my original work except for citations and quotations which have been duly acknowledged. I also declare that it has not been previously and concurrently submitted for any other degree or award at Xiamen University Malaysia or other institutions.

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APPROVAL FOR SUBMISSION

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ABSTRACT

In statistical inference, the frequentist approach treats parameters as unknown fixed constants, whereas the Bayesian framework views them as random variables following a probability distribution. While Monte Carlo methods allow for the computational approximation of these distributions, traditional centralized Markov Chain Monte Carlo (MCMC) becomes computationally prohibitive when dealing with large-scale or distributed datasets.

To address this, Consensus Monte Carlo offers a scalable alternative by partitioning data into shards, running separate MCMC chains locally, and combining the results to approximate the full posterior. This project implements a Consensus Monte Carlo framework using Independent Metropolis-Hastings (IMH) and Random Walk Metropolis-Hastings (RWMH). The study evaluates the accuracy of the consensus approach against a full-data Metropolis-Hastings baseline to determine the conditions under which the distributed method is an adequate substitute.

Keywords: Bayesian Inference, Markov Chain Monte Carlo (MCMC), Metropolis-Hastings, Consensus Monte Carlo.

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LIST OF SYMBOLS / ABBREVIATIONS

MCMC	: Markov Chain Monte Carlo
MH	: Metropolis–Hastings
RWMH	: Random Walk Metropolis–Hastings
IMH	: Independent Metropolis–Hastings
CMC	: Consensus Monte Carlo
β	: Regression coefficient vector
X	: Design matrix or predictor matrix
y	: Response variable
K	: Number of data subsets
$\pi(\beta \mid y, X)$	: Posterior distribution of β given y and X

CHAPTER 1

INTRODUCTION

1.1 Background Study

Bayesian inference is a statistical framework for learning about unknown quantities through probability. Let β denote a vector of unknown regression coefficients and let y denote the observed response variable. In the Bayesian approach, prior information about β is represented by a prior distribution $\pi(\beta)$, while the information provided by the observed data is represented by the likelihood function $L(y \mid \beta)$. These two components are combined through Bayes' theorem to obtain the posterior distribution,

$$\pi(\beta \mid y) = \frac{L(y \mid \beta)\pi(\beta)}{p(y)},$$

where $p(y)$ is the marginal likelihood. Since $p(y)$ does not depend on β , the posterior distribution is commonly written as

$$\pi(\beta \mid y) \propto L(y \mid \beta)\pi(\beta).$$

The posterior distribution is central in Bayesian inference because it describes the uncertainty about the unknown parameters after the data have been observed. Posterior means, posterior medians, posterior standard deviations, and credible intervals can be computed from this distribution when posterior samples or analytic expressions are available.

In many practical models, the posterior distribution cannot be evaluated directly in closed form. This difficulty often arises because the marginal likelihood involves integration over the parameter space, and the resulting posterior summaries may also require high-dimensional integration. Markov Chain Monte Carlo (MCMC) methods provide a computational approach for approximating posterior distributions by generating dependent samples from a Markov chain whose stationary distribution is the target posterior distribution. TODO: add citation for Bayesian inference and MCMC.

The Metropolis–Hastings algorithm is a general MCMC method for sampling from a target distribution that is known up to a proportionality constant. Starting from a current parameter value, the algorithm proposes a candidate value from a proposal distribution and then

accepts or rejects the candidate according to an acceptance probability determined by the target posterior density and the proposal mechanism. When the Markov chain is run for a sufficiently long period under suitable regularity conditions, the retained samples can be used to approximate posterior summaries. TODO: add citation for the Metropolis–Hastings algorithm.

For large datasets, full-data MCMC can become computationally expensive. If the data contain n observations, then each iteration of a full-data Metropolis–Hastings sampler may require evaluating the likelihood contribution of all n observations. As n increases, the cost of producing a sufficiently long Markov chain can become substantial. This issue is especially important when repeated likelihood evaluations are required for many proposed parameter values.

Distributed data settings provide one possible way to reduce this computational burden. Instead of treating the dataset as a single object processed by one sampler, the observations may be divided into K subsets. Let y_m denote the response data in subset m , where $m = 1, \dots, K$. Each subset can be processed separately, which allows the computational work to be divided across independent chains or computing resources.

Consensus Monte Carlo is a distributed Bayesian computation method based on this idea. The full posterior is approximated by first constructing subposteriors from the data subsets, running MCMC separately on each subset, and then combining the resulting subposterior samples to form an approximation to the full posterior distribution [Sco17, SBB⁺22]. Suppose the full dataset is partitioned into K subsets, $y_1, \dots, y_K$, and the likelihood can be factorised as

$$L(y \mid \beta) = \prod_{m=1}^K L(y_m \mid \beta).$$

In Consensus Monte Carlo, a common construction defines the m th subposterior as

$$\pi_m(\beta \mid y_m) \propto L(y_m \mid \beta)\pi(\beta)^{1/K},$$

so that the product of the subposteriors satisfies

$$\prod_{m=1}^K \pi_m(\beta \mid y_m) \propto \prod_{m=1}^K L(y_m \mid \beta)\pi(\beta)^{1/K} = L(y \mid \beta)\pi(\beta) \propto \pi(\beta \mid y).$$

Thus, the product of the subposteriors is proportional to the full posterior under this factorisation. In this project, Consensus Monte Carlo is studied together with the Metropolis–Hastings algorithm for distributed data.

The empirical application uses the UK accident dataset to illustrate Bayesian logistic regression within the distributed posterior simulation framework. Accident severity is suitable for this application because the response is binary after fatal and serious accidents are combined into a serious/fatal category and slight accidents are treated as the reference outcome. Logistic regression models the probability of serious/fatal accident severity relative to slight accident severity, while Bayesian inference provides posterior uncertainty for the regression coefficients. In the application, road type and speed limit are used as predictors of serious/fatal accident severity.

1.2 Problem Statement

Bayesian inference provides a coherent method for estimating unknown model parameters and quantifying uncertainty through the posterior distribution. However, its practical implementation can be computationally demanding when posterior simulation is required for a large dataset. In a full-data Metropolis–Hastings algorithm, the likelihood is commonly evaluated using all observations at each iteration. Therefore, the computational cost may increase considerably as the dataset becomes larger.

Consensus Monte Carlo reduces this computational burden by separating the full dataset into smaller subsets, running Metropolis–Hastings sampling on the corresponding subposteriors, and combining the resulting samples to approximate the full-data posterior. This approach is attractive because each subset chain uses only part of the data and can be treated as a separate computational task.

The main statistical issue is whether the consensus posterior is close to the posterior distribution obtained from full-data Metropolis–Hastings sampling. The approximation may be affected by the number of subsets, the way the data are partitioned, the prior adjustment used for the subposteriors, the behaviour of the subset Markov chains, and the posterior combination method. Consequently, it is necessary to compare the consensus posterior with the full-data posterior rather than assuming that the distributed method automatically gives equivalent results.

This project studies Consensus Monte Carlo using Metropolis–Hastings on distributed data, with Bayesian logistic regression for serious/fatal accident severity as the empirical application. The comparison focuses on posterior summaries, credible intervals, convergence behaviour, acceptance rates, and computational efficiency when runtime output is available. No claim about accuracy, runtime advantage, or practical superiority is made without support from the reported empirical results.

1.3 Research Objectives

The objectives of this project are as follows:

1. To formulate a Bayesian logistic regression model for accident severity, where the binary response represents serious/fatal accident severity relative to slight accident severity.
2. To implement full-data Metropolis–Hastings sampling for posterior estimation under the Bayesian logistic regression model.
3. To implement Consensus Monte Carlo by dividing the dataset into subsets and running Metropolis–Hastings separately on each subset subposterior.
4. To combine subposterior samples to approximate the full posterior distribution.
5. To compare full-data Metropolis–Hastings and Consensus Monte Carlo in terms of posterior summaries, credible intervals, convergence behaviour, acceptance rates, and computational runtime if available.

1.4 Research Framework

The research framework begins with data preprocessing. The UK accident dataset is inspected, cleaned, and arranged into a form suitable for Bayesian modelling. This stage includes constructing a binary accident severity response, where serious and fatal accidents are combined as the event category and slight accidents form the baseline category. The predictor information is represented through the design matrix X .

The second stage is Bayesian logistic regression model specification. Let y denote the binary response and let X denote the design matrix. The regression coefficient vector is

denoted by β , and the target distribution for the full-data analysis is the posterior distribution $\pi(\beta \mid y, X)$. This model provides the statistical basis for estimating how the selected predictors are associated with the probability of serious/fatal accident severity.

The third stage is full-data Metropolis–Hastings sampling. In this stage, a Markov chain is constructed to sample from the posterior distribution based on the complete dataset. The resulting samples provide the baseline posterior summaries for comparison with the distributed approach.

The fourth stage is data partitioning. The complete dataset is divided into K subsets, denoted by $y_1, \dots, y_K$ together with the corresponding predictor matrices. Each subset is used to define a subposterior distribution, with the prior contribution adjusted across subsets so that the product of the subposteriors corresponds to the full posterior structure.

The fifth stage is subset Metropolis–Hastings sampling. Separate Metropolis–Hastings chains are run on each subset to obtain subposterior samples. If β_m denotes a subset-specific parameter draw associated with subset m , then the collection of samples from all subsets is used as the input for the consensus step.

The sixth stage is consensus posterior combination. The subposterior samples are combined to construct an approximation to the full posterior distribution. This produces the consensus posterior sample used for comparison with the full-data Metropolis–Hastings sample.

The final stage is comparison. The full-data posterior and the consensus posterior are compared using posterior summaries, credible intervals, convergence behaviour, acceptance rates, and runtime values recorded during computation.

CHAPTER 2

LITERATURE REVIEW

2.1 A Weakly Informative Default Prior Distribution for Logistic and Other Regression Models

Gelman et al.'s paper addresses an important practical problem in logistic regression and related regression models: the instability of coefficient estimates when data are sparse, predictors are highly imbalanced, or separation occurs. In such cases, classical maximum likelihood estimation may produce very large or undefined coefficient estimates, making inference unreliable. The authors argue that fully informative priors may be difficult to specify in routine applied work, while completely noninformative priors may fail to regularise unstable likelihoods. Their contribution is therefore motivated by the need for a default Bayesian prior that is weak enough to allow the data to dominate when the data are informative, but strong enough to prevent implausibly large coefficient estimates in poorly identified logistic regression problems.

The main methodological contribution of the paper is the proposal of weakly informative Student- t prior distributions for coefficients in logistic and related regression models. In particular, the authors recommend a default Cauchy prior with centre 0 and scale 2.5 for regression coefficients after nonbinary predictors have been scaled appropriately. This paper is relevant to the present thesis because the methodology uses Bayesian logistic regression, and the choice of prior distribution affects posterior estimation before applying Metropolis–Hastings sampling and Consensus Monte Carlo on distributed data subsets. However, the paper is not primarily concerned with distributed Bayesian computation or posterior aggregation. Its role in this literature review is therefore to justify the Bayesian logistic regression and prior specification component of the methodology, rather than to provide a direct method for Consensus Monte Carlo. [GJPS08]

2.2 Comparing Consensus Monte Carlo Strategies for Distributed Bayesian Computation

This paper by Steven L. Scott reviews and compares methods for distributed Bayesian computation, focusing on how to combine posterior samples generated separately on different data partitions. Its main concern is the consensus Monte Carlo setting, where large datasets are split across machines to reduce computational burden and communication costs. Scott evaluates three main aggregation strategies already discussed in the literature: averaging methods, sequential resampling approaches, and density-estimation methods. The paper shows that simple weighted averaging performs very well when subposteriors are close to Gaussian, which often happens in large-sample settings, making it a strong practical baseline despite its simplicity.

A major contribution of the paper is its critical comparison of these methods and the proposal of a mixture-based consensus Monte Carlo approach, which uses Gaussian mixtures to better capture non-Gaussian posterior shapes. Scott finds that while more flexible methods such as kernel-based and sequential approaches are attractive in theory, they tend to perform poorly as dimension increases, whereas averaging remains stable and computationally efficient. The paper therefore concludes that in distributed Bayesian inference, the main challenge is not generating local posterior samples but combining them accurately without losing scalability. Overall, the study is important because it gives a balanced assessment of the trade-off between statistical accuracy and computational practicality in scalable Bayesian analysis. [Sco17]

2.3 Bayes and Big Data: The Consensus Monte Carlo Algorithm

Scott et al.’s Bayes and Big Data: The Consensus Monte Carlo Algorithm sits in the early literature on scalable Bayesian computation, addressing the problem of inference when data are too large for a single machine. The paper argues that existing parallel approaches based on multicore CPUs or GPUs reduce processor bottlenecks but do not solve memory and disk limits, while multi-machine settings introduce severe communication costs that make standard synchronized MCMC inefficient. In positioning their method, the authors connect their work to several adjacent strands: averaging frequentist estimators in distributed settings, the “bag of little bootstraps,” earlier Bayesian consensus ideas by Huang and Gelman,

and deterministic Bayesian approximations such as INLA and variational Bayes. Their literature framing is therefore important because it shows consensus Monte Carlo not as an isolated algorithm, but as one response within a broader effort to make inference feasible under big-data constraints while still preserving access to the full posterior distribution.

The paper’s main contribution is to propose consensus Monte Carlo, where data are split into shards, each worker runs an independent Monte Carlo simulation on its local subproblem, and the resulting draws are combined through averaging rules to approximate the full posterior. In the examples, the method performs well not only in Gaussian settings but also in some non-Gaussian cases, including beta-binomial models, hierarchical models, and BART, which supports the authors’ claim that simple averaging can be surprisingly effective in distributed Bayesian analysis. At the same time, the paper is careful about limitations: the method is best suited to fixed-dimensional continuous parameter spaces, and the authors explicitly note that performance away from Gaussianity remains an open question, with more sophisticated consensus mechanisms still needing development. As a result, this paper is best read as a foundational contribution that establishes the practicality of low-communication Bayesian aggregation, while also defining the main theoretical and methodological issues that later distributed MCMC research would need to resolve. [SBB⁺22]

2.4 Sampling for Bayesian computation with large datasets

Huang and Gelman’s Sampling for Bayesian Computation with Large Datasets is an early contribution to the literature on scalable Bayesian inference for hierarchical models. The paper begins from the observation that Bayesian multilevel modeling had already become important across fields such as health, education, environment, and business, but that standard Gibbs and related MCMC methods became difficult to use once datasets were too large for convenient storage and repeated full-data iteration. In that context, the authors position their work against earlier data-subsetting approaches, especially sequential and importance-sampling based methods such as Chopin (2002) and Ridgeway and Madigan (2002), arguing that their own strategy extends this line of work by using random partitioning and cluster sampling so that each subset preserves the hierarchical structure of the original data. This framing is important because it treats big-data Bayes not as a completely

new problem, but as an extension of existing work on approximate posterior computation under computational constraints.

The paper's main methodological contribution is a consensus-style posterior combination strategy: divide the data into disjoint subsets, fit the Bayesian model separately on each subset, and then combine the resulting inferences using a normal approximation, with possible refinement through importance sampling. In the literature, this paper is therefore best understood as a precursor to later consensus Monte Carlo methods, because it already identifies the central ideas that would define that area: subset-based inference, low communication cost, and the statistical problem of how to recombine local posterior information into a global approximation. The authors are also careful to state the unresolved issues, including the lack of a full theoretical justification, the choice of the number of subsets K , and the question of which aggregation rule is best. As a result, the paper is significant less for providing a final solution than for establishing the basic research agenda for distributed Bayesian computation in large hierarchical datasets. [HG05]

CHAPTER 3

METHODOLOGY

3.1 Data Source

The empirical analysis uses the UK accident dataset stored as `Dataset/accident_2013.csv`. TODO: add citation for the UK accident dataset source. The dataset contains accident-level information, including accident severity and road-related variables. The accompanying variable documentation is provided in `Dataset/Variables for UK data.pdf`.

The dataset is used as an applied setting for comparing full-data Metropolis–Hastings sampling with Consensus Monte Carlo. The statistical model is Bayesian logistic regression, and the response of interest is serious/fatal accident severity relative to slight accident severity. Thus, the accident data provide an empirical application for studying posterior simulation and posterior combination under a distributed-data framework.

3.2 Data Preprocessing

The empirical analysis uses three variables from the accident dataset: `road_type`, `speed_limit`, and `accident_severity`. Observations with missing values in these selected variables are removed using complete-case filtering. Observations for which `road_type` is recorded as `unknown` are also removed because this category does not represent an interpretable road type for the regression model.

The original accident severity variable is transformed into a binary response. Serious and fatal accidents are combined into the category `serious_fatal`, while slight accidents remain as the baseline category. Let y_i denote the binary response for observation i . In this study,

$$y_i = \begin{cases} 1, & \text{if observation } i \text{ is serious or fatal,} \\ 0, & \text{if observation } i \text{ is slight.} \end{cases}$$

Therefore, the event being modelled is serious/fatal accident severity relative to slight accident severity.

The predictors `road_type` and `speed_limit` are treated as categorical variables. The reference category for `road_type` is `single carriageway`, and the reference category for `speed_limit` is 2. A design matrix X is then constructed from the fitted logistic regression model, including the intercept and the indicator variables generated from the categorical predictors.

3.3 Bayesian Logistic Regression Model

Let $y_i \in \{0, 1\}$ be the binary accident severity response for observation i , and let x_i be the corresponding predictor vector, including the intercept term and the encoded categorical predictors. Let $\beta = (\beta_1, \dots, \beta_d)^T$ denote the vector of regression coefficients, where d is the number of regression coefficients in the model. The Bayesian logistic regression model assumes that

$$y_i \mid \beta \sim \text{Bernoulli}(p_i),$$

where p_i is the probability that observation i has serious/fatal accident severity. The logistic link function is

$$\text{logit}(p_i) = x_i^T \beta.$$

Equivalently,

$$p_i = \frac{\exp(x_i^T \beta)}{1 + \exp(x_i^T \beta)}.$$

For n independent observations, the likelihood function is

$$L(y \mid \beta, X) = \prod_{i=1}^n p_i^{y_i} (1 - p_i)^{1-y_i}.$$

Substituting the logistic form of p_i gives the log-likelihood

$$\log L(y \mid \beta, X) = \sum_{i=1}^n [y_i x_i^T \beta - \log\{1 + \exp(x_i^T \beta)\}].$$

The computation evaluates the term $\log\{1 + \exp(x_i^T \beta)\}$ using a numerically stable expression. TODO: add citation for Bayesian logistic regression.

3.4 Prior Distribution

An independent normal prior distribution is assigned to each regression coefficient. For coefficient β_j ,

$$\beta_j \sim N(0, \sigma_\beta^2),$$

where σ_β is the prior standard deviation. In this study, the prior standard deviation is set to

$$\sigma_\beta = 2.5.$$

This prior is centred at zero and provides regularisation for the logistic regression coefficients.

Under independence of the prior components, the prior density for β can be written as

$$\pi(\beta) = \prod_{j=1}^d \pi(\beta_j),$$

where d is the number of regression coefficients.

3.5 Posterior Distribution

Combining the likelihood and the prior distribution gives the posterior distribution of the regression coefficient vector:

$$\pi(\beta \mid y, X) \propto L(y \mid \beta, X)\pi(\beta).$$

The posterior distribution is sampled using Metropolis–Hastings algorithms because direct analytical posterior summaries are not used in this study. TODO: add citation for posterior simulation.

3.6 Full-Data Metropolis–Hastings Sampling

The full-data analysis uses the complete response vector y and design matrix X to sample from $\pi(\beta \mid y, X)$. Two Metropolis–Hastings approaches are considered: Random Walk Metropolis–Hastings and Independent Metropolis–Hastings. TODO: add citation for the Metropolis–Hastings algorithm.

Before the Bayesian simulation stage, a frequentist logistic regression model is fitted using the binary response and the selected predictors. Let $\hat{\beta}$ denote the estimated coefficient vector from this model, and let $\hat{\Sigma}$ denote the estimated covariance matrix obtained from `vcov(logistic_model)`. These quantities are used to construct starting values and proposal covariance matrices for the Metropolis–Hastings algorithms. Non-finite starting values are replaced by zero in the computation.

The computational procedure records runtime values for the Metropolis–Hastings methods using `Sys.time()` and stores them in a runtime table. These runtime values are used for computational comparison in Chapter 4, without affecting the posterior target distribution.

3.6.1 Random Walk Metropolis–Hastings Algorithm

In Random Walk Metropolis–Hastings, the proposal at iteration t is generated by perturbing the previous state. If $\beta^{(t-1)}$ is the current value, then the proposed value is

$$\beta^* = \beta^{(t-1)} + \epsilon,$$

where

$$\epsilon \sim N(0, \Sigma_{\text{RWMH}}).$$

For the full-data Random Walk Metropolis–Hastings sampler, the proposal covariance matrix is

$$\Sigma_{\text{RWMH}} = c_{\text{RWMH}}^2 \frac{2.38^2}{d} \hat{\Sigma},$$

where $c_{\text{RWMH}} = 0.75$, d is the number of regression coefficients, and $\hat{\Sigma}$ is the estimated covariance matrix from the fitted frequentist logistic regression model.

Since the random walk proposal is symmetric, the proposal densities cancel in the Metropolis–Hastings ratio. The acceptance probability is therefore

$$\alpha = \min \left\{ 1, \frac{\pi(\beta^* \mid y, X)}{\pi(\beta^{(t-1)} \mid y, X)} \right\}.$$

In log form, the proposal is accepted when

$$\log u < \log \pi(\beta^* \mid y, X) - \log \pi(\beta^{(t-1)} \mid y, X),$$

where $u \sim \text{Uniform}(0, 1)$.

3.6.2 Independent Metropolis–Hastings Algorithm

In Independent Metropolis–Hastings, the proposed value is generated from a fixed proposal distribution rather than by adding a random perturbation to the current state. The full-data Independent Metropolis–Hastings sampler uses

$$\beta^* \sim N(\hat{\beta}, \Sigma_{\text{IMH}}),$$

where

$$\Sigma_{\text{IMH}} = c_{\text{IMH}}^2 \hat{\Sigma}.$$

In this study, $c_{\text{IMH}} = 1.5$, $\hat{\beta}$ is the coefficient vector from the fitted frequentist logistic regression model, and $\hat{\Sigma}$ is the corresponding estimated covariance matrix.

Because the proposal distribution is not a symmetric random walk, the proposal-density correction is included in the acceptance probability. If the current value is $\beta^{(t-1)}$, then

$$\alpha = \min \left\{ 1, \frac{\pi(\beta^* | y, X) q(\beta^{(t-1)})}{\pi(\beta^{(t-1)} | y, X) q(\beta^*)} \right\}.$$

Equivalently, the log acceptance ratio is

$$\log \pi(\beta^* | y, X) - \log \pi(\beta^{(t-1)} | y, X) + \log q(\beta^{(t-1)}) - \log q(\beta^*).$$

3.7 Consensus Monte Carlo

Consensus Monte Carlo is used to approximate the full posterior distribution by dividing the data into subsets, sampling from subset-specific subposteriors, and combining the subposterior samples. TODO: add citation for Consensus Monte Carlo. This method is considered in order to study whether a distributed approximation can produce posterior summaries close to those from the full-data Metropolis–Hastings analysis.

3.7.1 Data Partitioning

Let the full dataset be partitioned into K subsets, where K denotes the number of data subsets. In this study, $K = 4$. The response vector and design matrix for subset m are denoted by y_m and X_m , respectively, where $m = 1, \dots, K$. Observations are randomly assigned to subsets before the subset samplers are run.

The full likelihood can be factorised across the subsets as

$$L(y \mid \beta, X) = \prod_{m=1}^K L(y_m \mid \beta, X_m).$$

3.7.2 Subposterior Distribution

For subset m , the subposterior distribution is defined by combining the subset likelihood with a fraction of the prior:

$$\pi_m(\beta \mid y_m, X_m) \propto L(y_m \mid \beta, X_m) \pi(\beta)^{1/K}.$$

The factor $\pi(\beta)^{1/K}$ distributes the prior contribution across the K subsets. This construction gives

$$\prod_{m=1}^K \pi_m(\beta \mid y_m, X_m) \propto \prod_{m=1}^K L(y_m \mid \beta, X_m) \pi(\beta)^{1/K} = L(y \mid \beta, X) \pi(\beta) \propto \pi(\beta \mid y, X).$$

Therefore, under the likelihood factorisation, the product of the subposterior densities is proportional to the full posterior density.

3.7.3 Subset Metropolis–Hastings Sampling

Separate Metropolis–Hastings chains are run on each subset to obtain samples from the subposterior distributions. The empirical analysis applies both subset Random Walk Metropolis–Hastings and subset Independent Metropolis–Hastings. For each subset, the target density in the acceptance probability is the subposterior density $\pi_m(\beta \mid y_m, X_m)$ rather than the full posterior density.

For subset Random Walk Metropolis–Hastings, the proposal is centred at the current value and uses a multivariate normal random walk proposal. The proposal covariance is based on the full-data covariance estimate and is multiplied by K to reflect the wider scale of the subset subposteriors:

$$\Sigma_{\text{CMC,RWMH}} = c_{\text{CMC,RWMH}}^2 \frac{2.38^2}{d} K \hat{\Sigma},$$

where $c_{\text{CMC,RWMH}} = 0.75$.

For subset Independent Metropolis–Hastings, a frequentist logistic regression model is fitted within each subset to obtain a subset-specific proposal mean and covariance matrix. If

$\hat{\beta}_m$ and $\hat{\Sigma}_m$ denote the subset-specific coefficient estimate and covariance estimate, then the proposal for subset m is

$$\beta^* \sim N(\hat{\beta}_m, \Sigma_{\text{CMC,IMH},m}),$$

where

$$\Sigma_{\text{CMC,IMH},m} = c_{\text{CMC,IMH}}^2 \hat{\Sigma}_m.$$

In this study, $c_{\text{CMC,IMH}} = 1.0$. Since this proposal is independent of the current state, the proposal-density correction is included in the Metropolis–Hastings acceptance probability.

After burn-in is removed, the retained subposterior samples from each subset are used in the consensus combination step. The number of retained samples is aligned across subsets by using the same available sample length from each subset chain.

3.7.4 Consensus Posterior Combination

The subposterior samples are combined using a precision-weighted consensus method. Let S denote the common number of retained posterior samples after burn-in, and let $\beta_m^{(s)}$ denote the s th retained sample from subset m , where $s = 1, \dots, S$. Let $\widehat{\text{Cov}}_m$ denote the estimated covariance matrix of the retained subposterior samples from subset m . The corresponding precision matrix is

$$W_m = \widehat{\text{Cov}}_m^{-1}.$$

The consensus sample at index s is defined as

$$\beta_{\text{consensus}}^{(s)} = \left(\sum_{m=1}^K W_m \right)^{-1} \sum_{m=1}^K W_m \beta_m^{(s)}.$$

This produces a sequence of consensus posterior samples

$$\{\beta_{\text{consensus}}^{(s)}\}_{s=1}^S,$$

which is summarised and compared with the full-data posterior samples.

3.8 Posterior Summary and Evaluation Criteria

The posterior samples obtained from full-data Metropolis–Hastings and Consensus Monte Carlo are evaluated using posterior summaries and diagnostic quantities. For each regression coefficient, the following summaries are considered:

1. posterior mean;
2. posterior median;
3. posterior standard deviation;
4. effective sample size;
5. Monte Carlo standard error;
6. 95% credible interval;
7. odds ratio;
8. 95% odds-ratio interval.

The empirical analysis estimates effective sample size using the autocorrelation structure of the sampled values and computes the Monte Carlo standard error as the posterior standard deviation divided by the square root of the estimated effective sample size. Odds ratios are obtained by exponentiating the posterior mean, and the odds-ratio interval is obtained by exponentiating the endpoints of the 95% credible interval.

Additional evaluation criteria include acceptance rate, trace plots, posterior histograms, and consensus posterior density plots. The acceptance rate is used to assess the behaviour of the Metropolis–Hastings proposal mechanism. Trace plots are used to visually inspect the mixing behaviour and stability of the chains. Posterior histograms and density plots are used to compare the marginal posterior distributions produced by the full-data and consensus approaches.

Computational efficiency is also considered through the runtime values recorded during the analysis. In Chapter 4, the comparison between full-data Metropolis–Hastings and Consensus Monte Carlo is based on posterior summaries, credible intervals, diagnostic measures, graphical checks, acceptance rates, and computational timing results.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Data Summary

This chapter reports the empirical results for Bayesian logistic regression applied to the UK accident dataset. The binary response is accident severity, where `serious_fatal` is coded as 1 and is the event being modelled, while `slight` is coded as 0 and is the reference response class. The predictors are `road_type` and `speed_limit`. The reference category for `road_type` is `single_carriageway`, and the reference category for `speed_limit` is 2.

Table 4.1 summarises the main analysis settings. After preprocessing, the analysis contains 82,344 observations and 10 regression coefficients. The prior standard deviation for the normal prior is 2.5. Both the full-data and subset Metropolis–Hastings samplers use 100,000 iterations, with a burn-in fraction of 0.2. Consensus Monte Carlo uses four subsets of equal size.

Table 4.1: Model settings for the empirical analysis

Setting	Value
Original observations	82570
Observations after preprocessing	82344
Number of coefficients	10
Prior standard deviation	2.5
Full-data MCMC iterations	100000
Full-data burn-in iterations	20000
Number of subsets, K	4
Subset MCMC iterations	100000
Subset burn-in iterations	20000

Table 4.2 shows the distribution of the binary response after preprocessing. Serious/fatal accidents account for 10,970 observations, corresponding to 13.32% of the analysis data.

Table 4.2: Distribution of the binary accident severity response

Response category	Count	Proportion
serious_fatal	10970	0.1332
slight	71374	0.8668

Tables 4.3 and 4.4 report the distributions of the categorical predictors. Single carriageway is the largest road-type category, and speed limit category 2 is the largest speed-limit category.

Table 4.3: Distribution of road type after preprocessing

Road type	Count	Proportion
dual carriageway	8439	0.1025
one way street (from 2005)	1720	0.0209
roundabout	9320	0.1132
single carriageway	61862	0.7513
slip road (from 2005)	1003	0.0122

Table 4.4: Distribution of speed limit after preprocessing

Speed limit category	Count	Proportion
1	1588	0.0193
2	61809	0.7506
3	6971	0.0847
4	2516	0.0306
5	7066	0.0858
6	2394	0.0291

4.2 Frequentist Logistic Regression Results

A frequentist logistic regression model is fitted before the Bayesian simulation stage. This model provides a baseline set of coefficient estimates and is also used to construct starting values and proposal covariance matrices for the Metropolis–Hastings algorithms. The frequentist results are not treated as the main inferential output of the thesis.

Table 4.5 reports the coefficient estimates, standard errors, and odds ratios from the frequentist logistic regression. The coefficients are on the log-odds scale for serious/fatal accident severity relative to slight accident severity.

Table 4.5: Frequentist logistic regression estimates and odds ratios

Parameter	Estimate	Std. Error	Odds Ratio
(Intercept)	-1.8981	0.0129	0.1499
road_typedual carriageway	-0.2250	0.0389	0.7985
road_typeone way street (from 2005)	-0.0109	0.0726	0.9891
road_typeroundabout	-0.4106	0.0368	0.6632
road_typeslip road (from 2005)	-0.7557	0.1174	0.4697
speed_limit1	0.1732	0.0717	1.1891
speed_limit3	0.1825	0.0383	1.2002
speed_limit4	0.3853	0.0569	1.4701
speed_limit5	0.5384	0.0327	1.7132
speed_limit6	0.3925	0.0684	1.4806

4.3 Full-Data Metropolis–Hastings Results

The full-data Bayesian analysis is performed using Random Walk Metropolis–Hastings and Independent Metropolis–Hastings. The posterior summaries include posterior mean, posterior median, posterior standard deviation, 95% credible interval, odds ratio, and 95% odds-ratio interval.

4.3.1 Random Walk Metropolis–Hastings

Table 4.6 reports selected posterior summaries for the full-data Random Walk Metropolis–Hastings sampler. The posterior mean and median are close for all coefficients. The 95% credible interval for `road_typeone way street (from 2005)` includes zero, while the intervals for the other reported non-intercept coefficients do not include zero.

Table 4.6: Full-data Random Walk Metropolis–Hastings posterior summaries

Parameter	Mean	Median	Std. Dev.	Lower 95% CrI	Upper 95% CrI	Odds Ratio	Lower 95 OR	Upper 95 OR
(Intercept)	-1.8981	-1.8982	0.0127	-1.9228	-1.8734	0.1498	0.1462	0.1536
road_typedual carriageway	-0.2250	-0.2251	0.0393	-0.3012	-0.1470	0.7985	0.7399	0.8633
road_typeone way street (from 2005)	-0.0123	-0.0111	0.0731	-0.1567	0.1284	0.9878	0.8550	1.1370
road_typeroundabout	-0.4114	-0.4114	0.0363	-0.4839	-0.3411	0.6627	0.6163	0.7110
road_typeslip road (from 2005)	-0.7574	-0.7543	0.1174	-0.9928	-0.5332	0.4689	0.3705	0.5867
speed_limit1	0.1703	0.1711	0.0713	0.0296	0.3069	1.1856	1.0300	1.3591
speed_limit3	0.1822	0.1826	0.0384	0.1065	0.2566	1.1999	1.1123	1.2926
speed_limit4	0.3834	0.3848	0.0569	0.2708	0.4907	1.4673	1.3111	1.6335
speed_limit5	0.5376	0.5380	0.0324	0.4738	0.6008	1.7119	1.6060	1.8236
speed_limit6	0.3892	0.3899	0.0683	0.2542	0.5219	1.4758	1.2894	1.6852

4.3.2 Independent Metropolis–Hastings

Table 4.7 reports the corresponding full-data Independent Metropolis–Hastings posterior summaries. The results are very close to the full-data Random Walk Metropolis–Hastings summaries, which indicates that the two full-data samplers produce similar posterior estimates for this analysis.

Table 4.7: Full-data Independent Metropolis–Hastings posterior summaries

Parameter	Mean	Median	Std. Dev.	Lower 95% CrI	Upper 95% CrI	Odds Ratio	Lower 95 OR	Upper 95 OR
(Intercept)	-1.8981	-1.8979	0.0132	-1.9236	-1.8723	0.1499	0.1461	0.1538
road_typedual carriageway	-0.2250	-0.2250	0.0391	-0.3025	-0.1490	0.7985	0.7390	0.8616
road_typeone way street (from 2005)	-0.0125	-0.0127	0.0732	-0.1583	0.1281	0.9875	0.8536	1.1367
road_typeroundabout	-0.4107	-0.4103	0.0366	-0.4835	-0.3403	0.6632	0.6166	0.7116
road_typeslip road (from 2005)	-0.7579	-0.7579	0.1170	-0.9922	-0.5322	0.4686	0.3708	0.5873
speed_limit1	0.1717	0.1729	0.0717	0.0285	0.3106	1.1873	1.0289	1.3643
speed_limit3	0.1817	0.1820	0.0383	0.1043	0.2567	1.1993	1.1099	1.2926
speed_limit4	0.3842	0.3854	0.0573	0.2700	0.4945	1.4685	1.3099	1.6397
speed_limit5	0.5379	0.5378	0.0330	0.4731	0.6027	1.7125	1.6049	1.8271
speed_limit6	0.3901	0.3899	0.0680	0.2572	0.5241	1.4772	1.2933	1.6890

4.4 Consensus Monte Carlo Results

Consensus Monte Carlo is implemented using four subsets, each containing 20,586 observations. The subset response distributions are similar across subsets: the numbers of serious/fatal observations are 2,743, 2,796, 2,676, and 2,755 for subsets 1 to 4, respectively. This provides a balanced partition for the reported distributed analysis.

4.4.1 Consensus Monte Carlo using Random Walk Metropolis–Hastings

Table 4.8 reports posterior summaries from Consensus Monte Carlo using subset Random Walk Metropolis–Hastings. The CMC-RWMH posterior means and credible intervals are generally close to the full-data posterior summaries.

Table 4.8: Consensus Monte Carlo using Random Walk Metropolis–Hastings posterior summaries

Parameter	Mean	Median	Std. Dev.	Lower 95% CrI	Upper 95% CrI	Odds Ratio	Lower 95 OR	Upper 95 OR
(Intercept)	-1.8978	-1.8977	0.0127	-1.9228	-1.8732	0.1499	0.1462	0.1536
road_typedual carriageway	-0.2255	-0.2256	0.0394	-0.3024	-0.1487	0.7981	0.7390	0.8618
road_typeone way street (from 2005)	-0.0109	-0.0107	0.0721	-0.1531	0.1305	0.9892	0.8581	1.1394
road_typeroundabout	-0.4145	-0.4145	0.0373	-0.4879	-0.3428	0.6607	0.6139	0.7098
road_typeslip road (from 2005)	-0.7793	-0.7783	0.1188	-1.0147	-0.5475	0.4587	0.3625	0.5784
speed_limit1	0.1772	0.1774	0.0727	0.0345	0.3191	1.1939	1.0351	1.3759
speed_limit3	0.1778	0.1778	0.0383	0.1034	0.2529	1.1945	1.1089	1.2878
speed_limit4	0.3856	0.3864	0.0569	0.2715	0.4960	1.4705	1.3120	1.6421
speed_limit5	0.5359	0.5360	0.0328	0.4715	0.6006	1.7091	1.6024	1.8231
speed_limit6	0.3885	0.3897	0.0702	0.2473	0.5239	1.4747	1.2806	1.6885

4.4.2 Consensus Monte Carlo using Independent Metropolis–Hastings

Table 4.9 reports posterior summaries from Consensus Monte Carlo using subset Independent Metropolis–Hastings. The CMC-IMH estimates are also close to the full-data estimates for most parameters. The acceptance rates reported later in this chapter are high for the CMC-IMH subset chains, so the numerical summaries should be interpreted together with the density plots and the comparison across methods.

Table 4.9: Consensus Monte Carlo using Independent Metropolis–Hastings posterior summaries

Parameter	Mean	Median	Std. Dev.	Lower 95% CrI	Upper 95% CrI	Odds Ratio	Lower 95 OR	Upper 95 OR
(Intercept)	-1.8982	-1.8982	0.0130	-1.9236	-1.8729	0.1498	0.1461	0.1537
road_typedual carriageway	-0.2260	-0.2259	0.0390	-0.3029	-0.1496	0.7977	0.7387	0.8610
road_typeone way street (from 2005)	-0.0088	-0.0083	0.0732	-0.1537	0.1328	0.9912	0.8575	1.1420
road_typeroundabout	-0.4109	-0.4108	0.0369	-0.4836	-0.3391	0.6630	0.6166	0.7124
road_typeslip road (from 2005)	-0.7731	-0.7710	0.1185	-1.0107	-0.5460	0.4616	0.3640	0.5793
speed_limit1	0.1793	0.1799	0.0722	0.0360	0.3194	1.1963	1.0366	1.3762
speed_limit3	0.1810	0.1812	0.0382	0.1056	0.2554	1.1984	1.1114	1.2910
speed_limit4	0.3884	0.3887	0.0570	0.2760	0.4989	1.4747	1.3178	1.6470
speed_limit5	0.5388	0.5388	0.0328	0.4744	0.6036	1.7140	1.6071	1.8286
speed_limit6	0.3934	0.3935	0.0686	0.2581	0.5269	1.4820	1.2944	1.6938

4.5 Comparison of Posterior Estimates

Table 4.10 compares posterior means across the four Bayesian methods. The two full-data samplers are very similar for all parameters. The CMC-RWMH and CMC-IMH posterior means are also close to the full-data posterior means for most coefficients. The more visible differences occur for `road_typeslip road (from 2005)` and `speed_limit1`, although the differences remain small on the log-odds scale.

Table 4.10: Comparison of posterior means across methods

Parameter	Full RWMH	Full IMH	CMC-RWMH	CMC-IMH
(Intercept)	-1.8981	-1.8981	-1.8978	-1.8982
road_typedual carriageway	-0.2250	-0.2250	-0.2255	-0.2260
road_typeone way street (from 2005)	-0.0123	-0.0125	-0.0109	-0.0088
road_typeroundabout	-0.4114	-0.4107	-0.4145	-0.4109
road_typeslip road (from 2005)	-0.7574	-0.7579	-0.7793	-0.7731
speed_limit1	0.1703	0.1717	0.1772	0.1793
speed_limit3	0.1822	0.1817	0.1778	0.1810
speed_limit4	0.3834	0.3842	0.3856	0.3884
speed_limit5	0.5376	0.5379	0.5359	0.5388
speed_limit6	0.3892	0.3901	0.3885	0.3934

The odds-ratio summaries show the same pattern because they are obtained by exponentiating the log-odds scale posterior summaries. Across the four methods, the 95% odds-ratio interval for `road_typeone way street (from 2005)` includes 1, while the corresponding intervals for the other non-intercept coefficients do not include 1. This is a descriptive posterior summary and should be interpreted in relation to the model specification, reference categories, and diagnostic results.

The effective sample size and Monte Carlo standard error outputs indicate that the full-data Independent Metropolis–Hastings sampler has larger effective sample sizes and smaller Monte Carlo standard errors than the full-data Random Walk Metropolis–Hastings sampler. The CMC-IMH effective sample size values are also large in the exported summaries. These values are not interpreted alone, because the Consensus Monte Carlo density plots are based on combined consensus samples rather than a single Markov chain, and diagnostic plots, acceptance rates, and runtime should be considered together.

4.6 Acceptance Rates and Computational Runtime

Table 4.11 reports the acceptance rates and runtime values. The full-data RWMH acceptance rate is 0.3922, while the full-data IMH acceptance rate is 0.2199. For Consensus Monte Carlo, the mean CMC-RWMH subset acceptance rate is 0.3943 and the mean CMC-IMH subset acceptance rate is 0.9046.

Table 4.11: Acceptance rates and computational runtime		
Method	Acceptance rate	Runtime minutes
Full-data RWMH	0.3922	19.30
Full-data IMH	0.2199	14.25
Mean CMC-RWMH	0.3943	17.18
Mean CMC-IMH	0.9046	18.43

The subset acceptance rates for CMC-RWMH are 0.3895, 0.3911, 0.4007, and 0.3960. The subset acceptance rates for CMC-IMH are 0.9071, 0.9050, 0.9025, and 0.9039. The runtime values indicate that, under the current sequential implementation, full-data IMH has the shortest recorded runtime, followed by CMC-RWMH, CMC-IMH, and full-data RWMH. These runtimes describe the present implementation and should not be interpreted as evidence of parallel speedup.

4.7 Diagnostic Plots

Figures 4.1 and 4.2 show posterior histograms and trace plots for the full-data Metropolis–Hastings samplers. These plots are used to visually assess the marginal posterior shapes and the stability of the sampled chains.

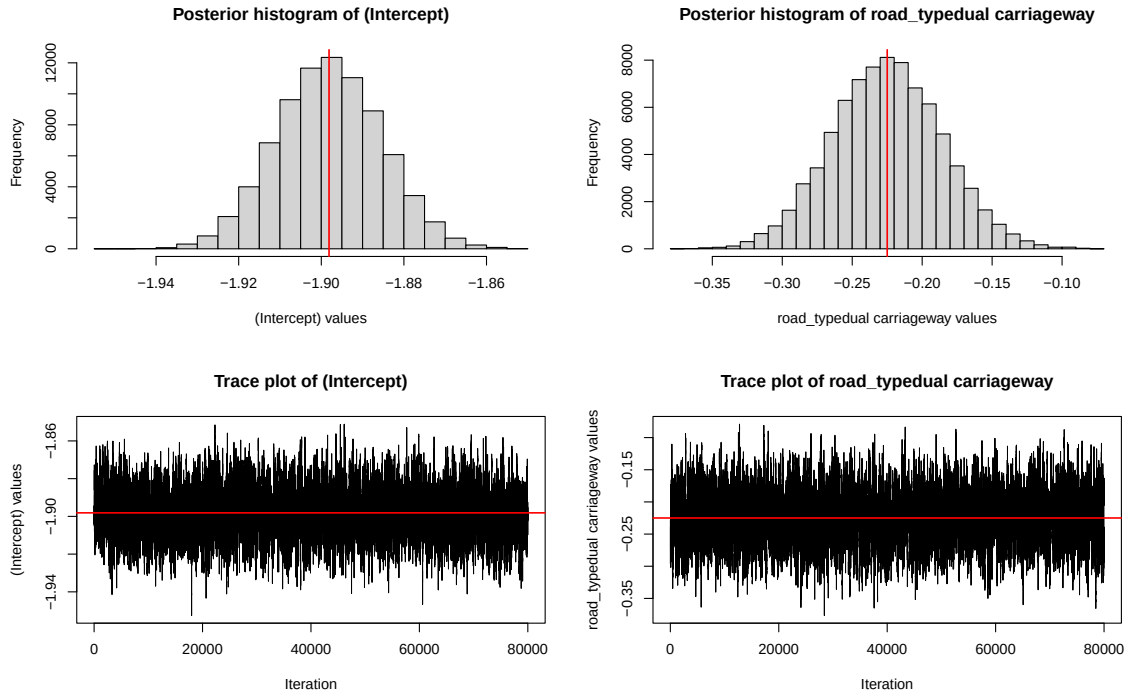


Figure 4.1: Full-data Random Walk Metropolis–Hastings posterior histograms and trace plots

Figures 4.3 and 4.4 show posterior density plots for the Consensus Monte Carlo results. These density plots are based on combined consensus samples and do not represent a single Markov chain. They are therefore used to compare marginal posterior shapes after posterior combination rather than to assess within-chain trace behaviour.

4.8 Summary of Findings

The empirical results show that the full-data Random Walk Metropolis–Hastings and full-data Independent Metropolis–Hastings samplers produce very similar posterior summaries for the Bayesian logistic regression model. This agreement provides a baseline for evaluating the Consensus Monte Carlo approximations.

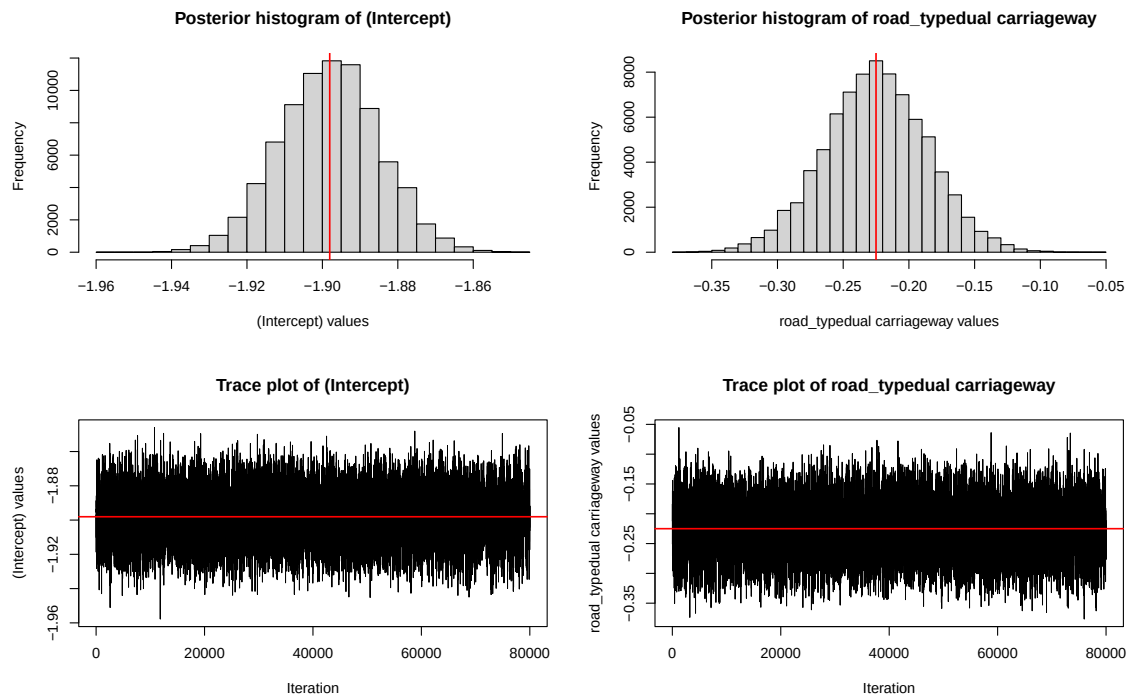


Figure 4.2: Full-data Independent Metropolis-Hastings posterior histograms and trace plots

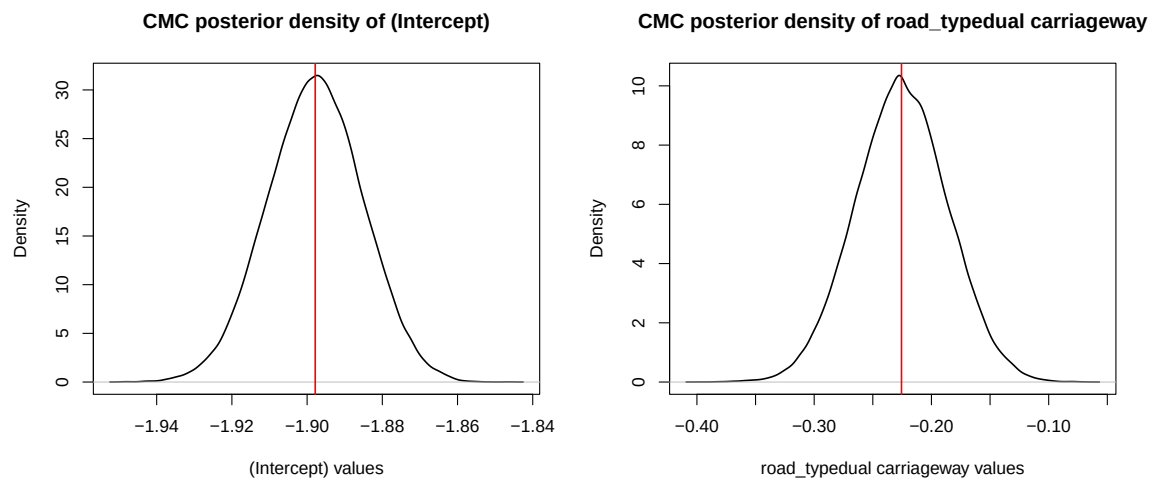


Figure 4.3: Consensus Monte Carlo using Random Walk Metropolis-Hastings posterior density plots

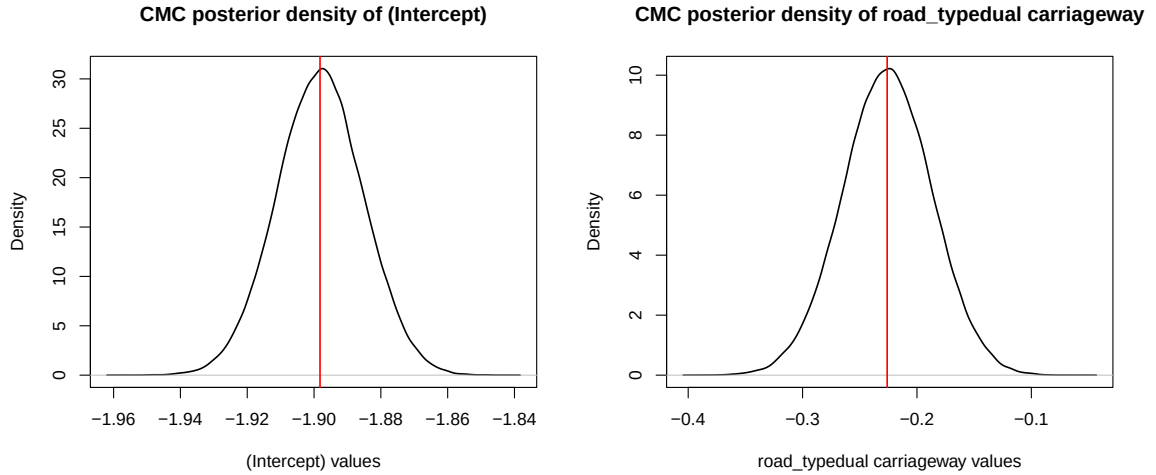


Figure 4.4: Consensus Monte Carlo using Independent Metropolis–Hastings posterior density plots

The Consensus Monte Carlo posterior summaries are generally close to the full-data posterior summaries. The CMC-RWMH and CMC-IMH posterior means differ most visibly from the full-data estimates for `road_typeslip road (from 2005)` and `speed_limit1`, but the overall posterior summaries remain similar across methods. The credible intervals and odds-ratio intervals also show broadly consistent conclusions across the four methods.

The acceptance-rate and runtime results should be interpreted descriptively. In the current sequential implementation, full-data IMH has the shortest recorded runtime, while CMC-RWMH and CMC-IMH do not demonstrate parallel speedup. Overall, the results suggest that Consensus Monte Carlo gives posterior summaries close to the full-data posterior summaries in this empirical analysis, but this conclusion is based on the reported posterior summaries, diagnostics, acceptance rates, and sequential runtime values rather than on an assumed computational advantage.

CHAPTER 5

CONCLUSION

APPENDIX A

APPENDIX

REFERENCES

- [GJPS08] Andrew Gelman, Aleks Jakulin, Maria Grazia Pittau, and Yu-Sung Su. A weakly informative default prior distribution for logistic and other regression models. *The Annals of Applied Statistics*, 2(4):1360–1383, 2008.
- [HG05] Zaijing Huang and Andrew Gelman. Sampling for bayesian computation with large datasets. *Available at SSRN 1010107*, 2005.
- [SBB⁺22] Steven L Scott, Alexander W Blocker, Fernando V Bonassi, Hugh A Chipman, Edward I George, and Robert E McCulloch. Bayes and big data: The consensus monte carlo algorithm. In *Big Data and Information Theory*, pages 8–18. Routledge, 2022.
- [Sco17] Steven L Scott. Comparing consensus monte carlo strategies for distributed bayesian computation. 2017.